AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM340
Natural Language Processing

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 15 / Apr /2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 120 minutes

Instructions:

- Total Marks: **60**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1

Multiple Choice Questions **(15 marks)**

Question 1, (15 marks, 1.5 mark for each question)
Answer the following MCQ's by selecting **the best** answer for each question. Write your **answer letter** in the answer sheet.

1. Which linguistic level focuses on how words are used in context to achieve specific goals?
   a) Phonology
   b) Syntax
   c) Pragmatics
   d) Morphology

2. Which Regex pattern matches "price: 100$" but NOT "price: 100"?
   a) price: \d+
   b) price: \d+\$
   c) price: \d*
   d) ^price: \d+

3. In the Porter Stemmer, the word "relational" is likely reduced to:
   a) relation
   b) relate
   c) relat
   d) relations

4. If characters at S[i] and T[j] are identical, the cost added to cell [i-1, j-1] is:
   a) 0
   b) 1
   c) 2
   d) -1

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5. "Sparsity" in Language Modeling refers to:
   a) Large training sets.
   b) Many zero-count N-grams in the test set.
   c) Small vocabularies.
   d) High-speed decoding.

6. Which smoothing method uses a heuristic parameter alpha (e.g., 0.4)?
   a) Laplace
   b) Good-Turing
   c) Stupid Backoff
   d) Kneser-Ney

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7. Which component of Naive Bayes represents the class probability before looking at features?
   a) Likelihood
   b) Posterior
   c) Prior
   d) Evidence

8. Naive Bayes is called "Naive" because it assumes:
   a) Features are dependent.
   b) Features are conditionally independent.
   c) The class is unknown.
   d) The text is normalized.

9. Logistic Regression is a "Discriminative" model because it:
   a) Learns P(d|c)
   b) Learns P(c|d)
   c) Ignores structure.
   d) Uses rules.

10. "Toxicity Detection" bias can lead to:
   a) Faster processing.
   b) Unfair censorship of minority dialects.
   c) Better accuracy.
   d) Increased memory usage.

PART 2

Short Essay Questions **(15 marks)**

Question 2, (15 marks)
Answer the following questions:

a) Define the Markov Assumption and explain why it is essential for calculating the probability of long word sequences in N-gram models. **[5 marks]**

b) Explain the difference between Word Tokenization and Sentence Segmentation. Why is the period character (".") particularly problematic for segmentation algorithms? **[5 marks]**

c) Contrast **Generative** and **Discriminative** classifiers. Why might a developer choose Logistic Regression over Naive Bayes for a dataset with highly correlated features? **[5 marks]**

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PART 3

Short problems/questions **(30 marks)**

Question 3, (10 marks)
Compute the Minimum Edit Distance (MED) matrix to transform **PLAY** into **STAY**:

- Costs: Insertion=1, Deletion=1, Substitution=2

You must:

a) Write the weighted distance matrix for transforming **PLAY** into **STAY**. **[6 marks]**

b) State the final Distance Score. **[2 marks]**

c) Identify the operation performed at the second character (L to T). **[2 marks]**

Question 4, (10 marks)
Given the following mini-corpus consisting of three sentences:

<s> the cat </s>
<s> the cat </s>
<s> the dog </s>

Answer the following **(show your calculations)**:

a) List the unique vocabulary V. **[2 marks]**

b) Calculate the Bigram Probability P(cat | the) using Laplace (Add-1) Smoothing.
[4 marks]

c) Calculate the Bigram Probability P(dog | the) using Laplace (Add-1) Smoothing.
[4 marks]

Question 5, (10 marks)
Assume a **Naive Bayes classifier** with the following known dataset:

Positive (+): great book
Positive (+): great price
Negative (-): bad book

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And given the document d = "**price book**",

Calculate the following using Laplace (Add-1) Smoothing **(show your calculations)**:

a) Prior probabilities P(+) and P(-)? **[2 marks]**

b) The Likelihoods for "price" and "book" for both classes? **[4 marks]**

c) The final Posterior scores? **[2 marks]**

d) What class will Naive Bayes assign to the document d? **[2 marks]**

End of questions

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